

DDE-SOLVER: A MAPLE PACKAGE FOR DISCRETE DIFFERENTIAL EQUATIONS

HADRIEN NOTARANTONIO, LIPN (PARIS, FRANCE)

ABSTRACT. We introduce the Maple package [DDE-Solver](#) for solving Discrete Differential Equations (DDEs) arising in enumerative combinatorics, typically in the enumeration of maps, trees, and walks. These equations are functional equations that algebraically relate a formal power series $F(t, u)$ to specializations of it with respect to the *catalytic* variable u . In 2006, Bousquet-Mélou and Jehanne proved that when these equations are of a particular form, then F is an algebraic series. [DDE-Solver](#) is based on their work and recent effective algorithms designed by Bostan, Safey El Din, and the author to compute an annihilating polynomial for $F(t, a)$, where a is a constant.

We give a table of running times of the implementation on known examples.

1. INTRODUCTION

Sequences of non-negative integers are ubiquitous in enumerative combinatorics. For instance, consider the set of bicoloured rooted planar maps (with faces either coloured in blue or in white, see Figure 1) such that the degree of each blue face is 3 and the degree of each white face is a multiple of 3. Such maps are called *3-constellations* (OEIS A069726). The sequence $(c_n)_{n \geq 0}$ that counts such maps with n blue faces starts with the integers

$$1, 1, 6, 54, 594, 7371, 99144, \dots$$

For such a sequence, many questions arise: does there exist a recurrence relation satisfied by $(c_n)_{n \geq 0}$? Does there exist a *closed formula* (i.e. an explicit formula) for $(c_n)_{n \geq 0}$? What is the value of c_{n_0} , for $n_0 \in \mathbb{Z}_{\geq 0}$ large (e.g. $n_0 = 3 \cdot 10^6$), and how fast can it be computed? What is the asymptotic growth of c_n when $n \rightarrow \infty$?

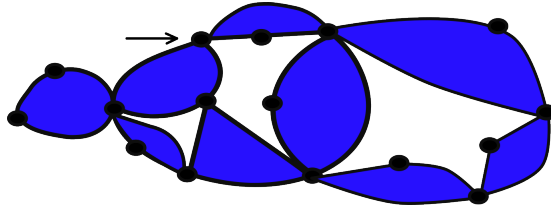


FIGURE 1. A 3-constellation contributing to the coefficient c_9 . Its white internal faces have either degree 3 or 6, and its external white face has degree 12. The arrow indicates the root vertex, and the orientation of the planar map is given by the edge following the root vertex, with the external face on its right.

To answer such questions, a common method is to introduce and study the properties of the associated generating function. For the above enumeration, one introduces $G(t) := \sum_{n \geq 0} c_n t^n \in \mathbb{Q}[[t]]$. For sophisticated enumerations, it is usually hard to study directly the generating function. To overcome these difficulties, one is often forced to refine the problem. This usually leads to introducing a new variable – called *catalytic* in what follows –, and to considering a bivariate generating function, say $F(t, u)$, where u is the catalytic variable. For well-chosen refinements, it is possible to write an equation that relates $F(t, u)$ and $G(t)$.

For the above example of 3-constellations, one typically refines the enumeration with the sequence $(c_{n,d})_{n,d \geq 0}$, where $c_{n,d}$ is the number of 3-constellations having n blue faces and outer degree $3d$. Naturally, one introduces its associated generating function $F(t, u) = \sum_{n,d \geq 0} c_{n,d} u^d t^n \in \mathbb{Q}[u][[t]]$. As $\sum_{d=0}^{\infty} c_{n,d} = c_n$, it holds that $F(t, 1) = G(t)$. After analyzing the construction of these objects (cf. [9, Fig. 7]), one deduces the equation

$$F(t, u) = 1 + tuF(t, u)^3 + tu \left(2F(t, u) + F(t, 1) \right) \frac{F(t, u) - F(t, 1)}{u - 1} + tu \frac{F(t, u) - F(t, 1) - (u - 1) \cdot \partial_u F(t, 1)}{(u - 1)^2}. \quad (1)$$

Now, for $a \in \mathbb{Q}$, denote $\Delta_a : F \in \mathbb{Q}[u][[t]] \mapsto \frac{F(t, u) - F(t, a)}{u - a} \in \mathbb{Q}[u][[t]]$ and Δ_a^ℓ the ℓ^{th} iteration of the *divided difference operator* Δ_a . When $\ell \geq 2$, observe that partial derivatives of F w.r.t. the variable u always appear in $\Delta_a^\ell(F(t, u))$: for instance, when $\ell = 2$, one has

$$\begin{aligned} \Delta_a^2(F(t, u)) &= \Delta_a \left(\frac{F(t, u) - F(t, a)}{u - a} \right), \\ &= \frac{\frac{F(t, u) - F(t, a)}{u - a} - \lim_{u \rightarrow a} \left(\frac{F(t, u) - F(t, a)}{u - a} \right)}{u - a}, \\ &= \frac{\frac{F(t, u) - F(t, a)}{u - a} - \partial_u F(t, a)}{u - a}, \\ &= \frac{F(t, u) - F(t, a) - (u - a) \cdot \partial_u F(t, a)}{(u - a)^2}. \end{aligned}$$

Therefore, (1) is of the form

$$F(t, u) = f(u) + t \cdot Q(F(t, u), \Delta_a F(t, u), \dots, \Delta_a^k F(t, u), t, u), \quad (2)$$

with $k > 0$ and f, Q polynomials (precisely with $k = 2$ and $a = 1$). Equations of the form (2) are called *discrete differential equations (DDEs)* of order k . Observe that one can derive a polynomial recurrence relation on the coefficients in t^n of a solution $F \in \mathbb{Q}[u][[t]]$ of (2), with an initial condition uniquely determined by $F(0, u) = f(u)$. Therefore, (2) has a unique solution in $\mathbb{Q}[u][[t]]$. Moreover, the following result proved by Bousquet-Mélou and Jehanne in [9], reminiscent of Popescu's theorem [24, Thm. 1.4], implies that this solution is annihilated by a nonzero polynomial with coefficients in $\mathbb{Q}(t, u)$.

Theorem 1. (cf. [9, Thm. 3]) *With the same notations as above, there exists a unique solution $F \in \mathbb{Q}[u][[t]]$ to the functional equation (2), and moreover $F(t, u)$ is algebraic over $\mathbb{Q}(t, u)$.*

Applying Theorem 1 to our guiding example, there exists some polynomial $R \in \mathbb{Q}[t, z] \setminus \{0\}$ such that $R(t, G(t)) = 0$. Namely, $G(t) = 1 + t + 6t^2 + 54t^3 + 594t^4 + \dots$ is annihilated by

$$R(t, z) = 81t^2z^3 - 9t(9t - 2)z^2 + (27t^2 - 66t + 1)z - 3t^2 + 47t - 1.$$

Moreover G is the unique root of R in $\mathbb{Q}[[t]]$. Thus R , together with the initial condition $R(0, c_0) = 0$, provides a finite amount of data (its monomials) that encodes the sequence $(c_n)_{n \geq 0}$. Now, using R , we write the linear differential equation satisfied by $F(t, 1)$ (for instance by using the Maple package `gfun` [25] by Salvy and Zimmermann) and deduce (still with `gfun`) the associated recurrence relation:

$$(81n^2 + 81n + 18) \cdot c_n - (4n^2 + 14n + 12) \cdot c_{n+1} = 0,$$

with $c_0 = 1, c_1 = 1$. Using this recurrence relation, computing $c_{3 \cdot 10^6}$ only takes 5 seconds in Maple. Solving the recurrence relation, one obtains $c_0 = 1$ and $c_n = \frac{4 \cdot 3^{n-1}}{(2n+2)(2n+1)} \binom{3n}{n}$, for all $n \geq 1$ (OEIS A069726). This *closed formula* yields the asymptotic behavior

$$c_n \sim \frac{3^{4n} 2^{-2n}}{2\sqrt{3\pi}} \left(n^{-\frac{5}{2}} + O(n^{-\frac{7}{2}}) \right), \text{ when } n \rightarrow \infty.$$

The goal of **DDE-Solver** is to automatically perform the following task:

For a solution $F(t, u)$ of a DDE of the form (2), compute $R \in \mathbb{Q}[t, z] \setminus \{0\}$ such that $R(t, F(t, a)) = 0$.

Previous algorithmic works. There exists a rich literature regarding the effective resolution of DDEs. The articles [6, 7, 9] already contain a complete account of the state of the art, so we only give a quick summary of this literature. Let us start with DDEs such that the polynomial Q in (2) has degree 2 in its first variable. When $k = 1$ (cf. (2)), Brown introduced in [14] the *quadratic method*, which was generalized later by Bender and Canfield in [3] to a particular family of DDEs of arbitrary order, still with Q of degree 2 in its first variable.

Equations in which Q is linear in all its arguments (except t, u) appear in many walk-enumeration problems (e.g. [9, §3.1]). Solving linear DDEs is done by applying the *kernel method*, introduced by Knuth in [21] (as pointed out in [1]) and extended by Bousquet-Mélou and Petkovšek in [11].

Strategies based on the *guess-and-prove* paradigm (popularized in [23]) have been studied by Zeilberger [26] when $k = 1$ (cf. (2)), and improved, in this same case, by Gessel and Zeilberger in [20].

Recall that k denotes the order of the DDE. Bousquet-Mélou and Jehanne designed in [9] a general method which can be seen as a generalization of both the kernel and the quadratic methods. Their strategy consists in translating the resolution of a DDE into the resolution of a polynomial system over $\mathbb{Q}(t)$, in $3k$ equations and $3k$ unknowns, with one of its solutions admitting $F(t, a)$ as one of its coordinates. The main step then consists of eliminating all variables except the one that stands for $F(t, a)$. We refer to their algorithm as “duplication”.

Recently, intensive studies have been carried out in [6] (resp. in [7]) in the direction of effectivity to design efficient algorithms for solving DDEs of order $k = 1$ (resp. of any order). The authors design three algorithms, referred to as “elimination”, “geometry” and “hybrid”.

The first two algorithms each rely on a distinct geometric observation that interprets the series $F(t, a)$ as the coordinate of a point α obtained by projecting another point, where the preimage of α under the associated projections has a cardinality greater than or equal to k .

The third algorithm, known as the “hybrid” algorithm, precomputes upper bounds on the partial degrees of the minimal polynomial of $F(t, a)$, then computes a candidate polynomial for annihilating $F(t, a)$, then proves that this guess is correct by checking that *enough* terms are annihilated. This last step is done by relying on a *zero lemma* that (i) makes explicit a positive integer N that depends on the precomputed upper bounds, (ii) proves that if $F(t, a) \bmod t^N$ is annihilated by this guessed polynomial, then it in fact annihilates $F(t, a)$.

Contributions. This paper contains one main contribution. It describes the Maple package **DDE-Solver**, dedicated to solving DDEs of any order that satisfy the (harmless) assumptions made explicit in [7]. Explicitly, **DDE-Solver** contains the function

algebraic_single.

This function takes as input a DDE and outputs an annihilating polynomial using one of the algorithms “duplication”, “elimination”, “geometry” or “hybrid”, depending on the user’s choice. Let us emphasize that **DDE-Solver** will grow and contain many other functions in the future.

Unlike [7], we have added four new DDEs to the execution time table. These new DDEs include a new family of DDEs, namely m -greedy Tamari intervals [13] (where m is a parameter that increases with the order of the DDEs). In particular, when $m = 3$, this yields the solving of the associated DDE in a few seconds, when the algorithm “duplication” does not terminate in less than a week.

Structure of the paper. In the preliminaries section (Section 2), we first recall the link between DDEs and polynomial systems; we then state some preliminaries on effective algebraic geometry that are used in Section 3 when giving an overview on how the algorithms “duplication”, “elimination”, “geometry” and “hybrid” (resp. [7, §3, §5, §6, §4]) work. In Section 4, we explain the computational paradigm on which the implementations are based, then how to get started with **DDE-Solver**, and how to use the options of **algebraic_single**. Ultimately, Section 4 contains some timings that illustrate the efficiency of this new package on various DDEs from the literature [4, 8, 9, 13].

Notations. Denote by $\mathbb{K}$ a field of characteristic 0, by $\overline{\mathbb{K}}$ its algebraic closure, by $\mathbb{Q}$ the field of rational numbers and by $\mathbb{F}_p$ the finite field with p elements. For a fixed positive integer k , denote by $\mathbf{x}$ (resp. $\mathbf{u}$ and $\mathbf{z}$) the variables $x_1, \dots, x_k$ (resp. $u_1, \dots, u_k$ and $z_0, \dots, z_{k-1}$). Denote by $\mathbb{K}[\mathbf{x}]$ the ring of polynomials in the variables $\mathbf{x}$ with coefficients in $\mathbb{K}$. For $P \in \mathbb{K}[\mathbf{x}]$, denote by $\mathbf{V}(P) \subset \overline{\mathbb{K}}^k$ its zero set. For any ideal $\mathcal{I} \subset \mathbb{K}[\mathbf{x}]$ and any set of polynomials $\mathcal{S} \subset \mathbb{K}[\mathbf{x}]$, denote by $\mathbf{V}(\mathcal{I}) \subset \overline{\mathbb{K}}^k$ (resp. by $\mathbf{V}(\mathcal{S}) \subset \overline{\mathbb{K}}^k$) the zero set of $\mathcal{I}$ (resp. of $\mathcal{S}$). Still for $P \in \mathbb{K}[\mathbf{x}]$, denote by $\text{disc}_{x_j}(P)$ the discriminant [16, Ex. 16, §6, Ch. 3] of P with respect to x_j and by $\text{LC}_{x_j}(P)$ the leading coefficient of P with respect to x_j . Denote by $\overline{\mathbb{K}}[[t^{\frac{1}{d}}]]$ the ring $\bigcup_{d \geq 1} \overline{\mathbb{K}}[[t^{\frac{1}{d}}]]$ of Puiseux series in t with nonnegative rational exponents (and bounded denominator).

2. PRELIMINARIES

2.1. FROM DDES TO POLYNOMIAL SYSTEMS

Starting from a DDE of the form (2), we multiply it by the least power of $(u - a)$ such that we obtain a polynomial functional equation of the form

$$P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \quad (3)$$

for some polynomial $P \in \mathbb{Q}[x, z, t, u]$. By the chain rule, taking the derivative of (3) with respect to u yields

$$\begin{aligned} \partial_u F(t, u) &\cdot \partial_x P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) \\ &+ \partial_u P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0. \end{aligned} \quad (4)$$

Consider the equation in u given by

$$\partial_x P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0. \quad (5)$$

Observe that the solutions of (5) that belong to $\overline{\mathbb{Q}}[[t^{\frac{1}{*}}]]$ are also solutions, by using (4), of the equation in u given by

$$\partial_u P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0.$$

Thus any non-constant solution in $\overline{\mathbb{Q}}[[t^{\frac{1}{*}}]]$ of (5) is a solution in u of the system of constraints

$$\begin{cases} P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \\ \partial_x P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, & u(u - a) \neq 0, \\ \partial_u P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0. \end{cases} \quad (6)$$

All the algorithms of the paper rely on the following hypothesis. This assumption holds for the examples considered in this paper.

Global assumption: There exist k distinct solutions $U_1(t), \dots, U_k(t) \in \overline{\mathbb{Q}}[[t^{\frac{1}{*}}]]$ (GA) to the system of equations (6). Also, $U_1(t), \dots, U_k(t) \notin \overline{\mathbb{Q}}$.

It follows from [9, Thm. 3] that the series $\{\partial_u^{i-1} F(t, a), U_i(t), F(t, U_i(t))\}_{1 \leq i \leq k}$ are in $\overline{\mathbb{Q}(t)}$. In Section 3, we show how (GA) allows one to design the algorithms implemented in DDE-Solver.

However, let us mention that there exist examples in the literature for which (GA) is not satisfied. For instance, in [9, Eq. 42], the authors consider an equation with one catalytic variable of order $k = 2$ such that there exist two solutions $U_1, U_2 \in \overline{\mathbb{Q}}[[t^{\frac{1}{*}}]]$ to the system (6). Unfortunately, these solutions are not distinct. To overcome this situation, the common strategy is to introduce a deformation of the corresponding DDE. This deformation is detailed in the proof of [9, Thm. 3] and will be implemented in a future version to overcome this limitation.

2.2. BRIEF OVERVIEW ON POLYNOMIAL TOOLS

This section introduces the tools used in Section 3. Most theorems and facts in this section are taken or deduced from [16, 17].

2.2.1. Algebraic elimination via Gröbner bases. For $\alpha = (\alpha_1, \dots, \alpha_k) \in \mathbb{Z}_{\geq 0}^k$, denote $\mathbf{x}^\alpha := x_1^{\alpha_1} \cdots x_k^{\alpha_k}$. For $S \subset \mathbb{K}[\mathbf{x}]$, denote by $\langle S \rangle$ the ideal generated by S in $\mathbb{K}[\mathbf{x}]$. We refer to [16, Chap. 2] for the definitions of monomial order, leading monomial and leading term. For a monomial order $\succ$ on $\mathbb{K}[\mathbf{x}]$ and a polynomial $p \in \mathbb{K}[\mathbf{x}]$, we denote by $\text{LT}_\succ(p)$ the leading term of p with respect to $\succ$.

Definition 1. (cf. [16, Chap. 2, §5, Def. 5]) Fix a monomial order $\succ$ on the ring $\mathbb{K}[\mathbf{x}]$. A finite subset $G = \{g_1, \dots, g_s\}$ of a nonzero ideal $\mathcal{I} \subset \mathbb{K}[\mathbf{x}]$ is said to be a Gröbner basis of $\mathcal{I}$ for the order $\succ$ if the equality $\langle \text{LT}(g_1), \dots, \text{LT}(g_s) \rangle = \langle \{\text{LT}(g) \mid g \in \mathcal{I}\} \rangle$ of monomial ideals holds.

By [16, Chap. 2, §5, Cor. 6], every nonzero ideal $\mathcal{I}$ admits a Gröbner basis and any Gröbner basis of $\mathcal{I}$ generates $\mathcal{I}$.

Gröbner bases and projections. The results below justify the use of Gröbner bases as a dedicated theoretical and computational tool, allowing one to characterize projections as solution sets of conjunctions of polynomial equations and inequations.

Denote by $\succ_{\text{lex}}$ the lexicographic order on $\mathbb{K}[x]$ with $x_1 > x_2 > \dots > x_k$.

Theorem 2 (Elimination theorem). (cf. [16, Chap. 3, §1, Thm. 2]) *Let $\mathcal{I} \subset \mathbb{K}[x]$ be an ideal. Denote by G a Gröbner basis of $\mathcal{I}$ with respect to the order $\succ_{\text{lex}}$. Then for every $0 \leq \ell \leq k-1$, the set $G_\ell = G \cap \mathbb{K}[x_{\ell+1}, \dots, x_k]$ is a Gröbner basis of the ideal $\mathcal{I}_\ell := \mathcal{I} \cap \mathbb{K}[x_{\ell+1}, \dots, x_k]$.*

By Theorem 2, the set of polynomials G_ℓ finitely generates $\mathcal{I}_\ell$. Now, denote the geometric projection $\pi_\ell : (x_1, \dots, x_k) \in \overline{\mathbb{K}}^k \mapsto (x_{\ell+1}, \dots, x_k) \in \overline{\mathbb{K}}^{k-\ell}$. Recall that a Zariski closed set in $\overline{\mathbb{K}}^k$ is defined as the solution set of some polynomial equations defined in $\mathbb{K}[x]$.

Theorem 3 (Closure theorem). (cf. [16, Chap. 3, §2, Thm. 3]) *Let $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_k]$ be an ideal and $\mathbf{V}(\mathcal{I}) \subset \overline{\mathbb{K}}^k$ be its zero set. Consider $\mathcal{I}_\ell := \mathcal{I} \cap \mathbb{K}[x_{\ell+1}, \dots, x_k]$. Then $\mathbf{V}(\mathcal{I}_\ell)$ is the smallest Zariski closed set containing $\pi_\ell(\mathbf{V}(\mathcal{I})) \subset \overline{\mathbb{K}}^{k-\ell}$.*

Theorems 2 and 3 imply that characterizing the Zariski closure of the projection of a zero set $\mathbf{V}(\mathcal{I})$ onto some coordinate subspace is done by computing a Gröbner basis for $\succ_{\text{lex}}$.

Theorem 4 (Extension theorem). (cf. [16, Chap. 3, §5, Thm. 2]) *Let $G = \{g_1, \dots, g_s\}$ be a Gröbner basis of $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_k]$ for the order $\succ_{\text{lex}}$. For each $1 \leq j \leq s$, consider*

$$g_j = c_j(x_2, \dots, x_k) \cdot x_1^{N_j} + (\text{terms in which } x_1 \text{ has degree } < N_j),$$

where $N_j \geq 0$ and $c_j \in \mathbb{K}[x_2, \dots, x_k]$ is nonzero. Assume $\alpha = (\alpha_2, \dots, \alpha_k) \in \mathbf{V}(\mathcal{I} \cap \mathbb{K}[x_2, \dots, x_k])$ is a partial solution with the property that $\alpha \notin \mathbf{V}(c_1, \dots, c_s)$. Then

$$\langle f(x_1, \alpha) \mid f \in \mathcal{I} \rangle = \langle g_{j_0}(x_1, \alpha) \rangle,$$

where $g_{j_0} \in G$ satisfies $c_{j_0}(\alpha) \neq 0$ and g_{j_0} has minimal degree in x_1 among all elements g_j with $c_j(\alpha) \neq 0$. Furthermore, if $g_{j_0}(\alpha_1, \alpha) = 0$ for $\alpha_1 \in \overline{\mathbb{K}}$, then $(\alpha_1, \alpha) \in \mathbf{V}(\mathcal{I})$.

Theorem 4 implies that characterizing projections (and not their Zariski closure) is done by considering the disjunction of inequations associated with the non-simultaneous vanishing of the $c_1, \dots, c_s$ (notations of Theorem 4): for each condition in this disjunction, one shall add the vanishing of all the elements of G_ℓ in Theorem 2.

2.2.2. Counting specific solutions. Let $s \geq 1$ be an integer. Denote $\mathbf{y} = y_1, \dots, y_s$. Let $\mathcal{I} \subset \mathbb{K}[x]$ be a radical ideal such that its zero set $\mathbf{V}(\mathcal{I}) \subset \overline{\mathbb{K}}^k$ is finite. For $\alpha \in \mathbf{V}(\mathcal{I})$, denote by $[x_1](\alpha)$ the x_1 -coordinate of α . Denote the projection $\pi_{x_1} : (x) \in \overline{\mathbb{K}}^k \mapsto x_1 \in \overline{\mathbb{K}}$. Let $g \in \mathbb{K}[\mathbf{y}][z]$ and denote $\text{LC}_z(g) \in \mathbb{K}[\mathbf{y}]$ the leading coefficient of g in z .

Problem 5. *Let $\ell \geq 1$ be an integer. Characterize in $\mathbb{K}[x]$ the set*

$$\{\beta \in \mathbf{V}(\mathcal{I}) \mid \#(\pi_{x_1}^{-1}([x_1](\beta)) \cap \mathbf{V}(\mathcal{I})) \geq \ell\}.$$

Problem 6. *Let $1 \leq \ell \leq \deg_z(g)$. Characterize in $\mathbb{K}[\mathbf{y}]$ the points $\beta \in \overline{\mathbb{K}}^s \setminus \mathbf{V}(\text{LC}_z(g))$ such that $g(\mathbf{y} = \beta, z)$ has at least ℓ distinct roots.*

Answer to Problem 5. Let $\mathcal{A} := \mathbb{K}[x]/\mathcal{I}$. It results from the finiteness of $\mathbf{V}(\mathcal{I})$ that $\mathcal{A}$ has finite dimension as a $\mathbb{K}$ -vector space (cf. [16, Chap. 5, §3, Thm. 6]). Let $m_{x_1} : f \in \mathcal{A} \mapsto x_1 \cdot f \in \mathcal{A}$ be the *multiplication map by x_1* in $\mathcal{A}$. Let $\chi_{x_1} \in \mathbb{K}[T]$ be its characteristic polynomial. The following is the content of [17, Chap. 4, §2, Prop. 2.7], where the exponents in the product are equal to 1 as $\mathcal{I}$ is a radical ideal (cf. [17, Chap. 4, §2, Prop. 2.6]).

Fact 1. We have the equality $\chi_{x_1} = \pm \prod_{\alpha \in \mathbf{V}(\mathcal{I})} (T - [x_1](\alpha))$.

As a consequence of Fact 1, we obtain the following equality of sets:

$$\begin{aligned} \{\beta \in \mathbf{V}(\mathcal{I}) \mid \#(\pi_{x_1}^{-1}([x_1](\beta)) \cap \mathbf{V}(\mathcal{I})) \geq \ell\} \\ = \{\beta \in \overline{\mathbb{K}}^k \mid \beta \in \mathbf{V}(\mathcal{I}) \wedge \chi_{x_1}([x_1](\beta)) = 0 \wedge \cdots \wedge \partial_T^{\ell-1} \chi_{x_1}([x_1](\beta)) = 0\}. \end{aligned} \quad (7)$$

Eq. (7) provides a conjunction of polynomial equations that answers Problem 5.

Answer to Problem 6. Define the *Hermite quadratic form* associated with g by

$$H_g : (f, h) \in (\mathbb{K}(\mathbf{y})[z]/\langle g \rangle)^2 \mapsto \text{Trace}(m_{f \cdot h}) \in \mathbb{K}(\mathbf{y}),$$

where $\text{Trace}(\cdot)$ is the trace operator and $m_{f \cdot h}$ is the multiplication map by $f \cdot h$ in $\mathbb{K}(\mathbf{y})[z]/\langle g \rangle$. Also, denote by $M_{H_g} \in \mathbb{K}(\mathbf{y})^{\deg_z(g) \times \deg_z(g)}$ the matrix of H_g in the basis $\{1, z, \dots, z^{\deg_z(g)-1}\}$.

The following fact is an immediate consequence of [17, Chap. 2, §5, Thm. 5.2] and of the observation that, by performing Euclidean divisions by g in $\mathbb{K}(\mathbf{y})[z]$, the denominator of the image of z^i in $\mathbb{K}(\mathbf{y})[z]/\langle g \rangle$ can only be a power of $\text{LC}_z(g)$.

Fact 2. The denominators in the matrix M_{H_g} are powers of $\text{LC}_z(g)$. Moreover if $1 \leq \ell \leq \deg_z(g)$, then the points $\beta \in \overline{\mathbb{K}}^s \setminus \mathbf{V}(\text{LC}_z(g))$ at which $g(\mathbf{y} = \beta, z)$ has fewer than ℓ distinct solutions are the points $\beta \in \overline{\mathbb{K}}^s \setminus \mathbf{V}(\text{LC}_z(g))$ at which the $\ell \times \ell$ -minors of M_{H_g} all vanish.

Fact 2 implies that the disjunction of polynomial inequations answering Problem 6 is given by the non-vanishing of $\text{LC}_z(g)$ and by the non-vanishing of at least one $\ell \times \ell$ -minors of M_{H_g} .

2.3. CHANGE OF MONOMIAL ORDERING

For k generic elements of $\mathbb{K}[x]$ of degree d , computing a Gröbner basis of $\mathcal{I}$ for a graded monomial order, for instance $\succ_{\text{grevlex}}$ (cf. [16, Chap. 2, §2, Def. 6]), has an arithmetic cost which is in $d^{O(k)}$ (cf. [2, Prop. 1]).

Let $\mathcal{I} \subset \mathbb{K}[x]$ be an ideal whose solution set is finite in $\overline{\mathbb{K}}^k$. The following fact uses the FGLM (cf. [18]) algorithm which, on input a Gröbner basis of $\mathcal{I}$ for a graded monomial ordering like $\succ_{\text{grevlex}}$, outputs a Gröbner basis of $\mathcal{I}$ for a lexicographic monomial ordering (e.g. $\succ_{\text{lex}}$ from Section 2.2.1). The complexity of this algorithm is, under some genericity assumptions (e.g. $\mathcal{I}$ radical), in $\tilde{O}(D^\omega)^1$ (cf. [22]), where D is the cardinality of $\mathbf{V}(\mathcal{I})$. By the Bézout bound, and with the notations of the above paragraph, we have $D \leq d^k$, this makes the below fact useful for computing in $d^{O(k)}$ a Gröbner basis of $\mathcal{I}$ for a lexicographic order.

Fact 3. Let $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_k]$ be an ideal whose solution set $\mathbf{V}(\mathcal{I}) \subset \overline{\mathbb{K}}^k$ is finite. There exists an algorithm that computes a Gröbner basis $G_{\succ_{\text{grevlex}}}$ of $\mathcal{I}$ for a graded monomial order and that turns it into a Gröbner basis $G_{\succ_{\text{lex}}}$ of $\mathcal{I}$ for a lexicographic order.

¹We denote by ω the exponent of multiplication matrix, that is, multiplying two matrices in $\mathbb{K}^{k \times k}$ can be done by using $\tilde{O}(k^\omega)$ arithmetic operations in $\mathbb{K}$, where $\tilde{O}(\cdot)$ is a $O(\cdot)$ that hides polylogarithmic factors.

With the notations as above and without using the FGLM algorithm of [18], the computation of a Gröbner basis of $\mathcal{I}$ for a lexicographic order has an arithmetic cost bounded by $C_1 d^{C_2 k^3}$ [15], for $C_1, C_2 \in \mathbb{Z}_{>0}$.

3. DETAILS ON THE ALGORITHMS

As explained in the next section, currently, the only function implemented in **DDE-Solver** is: **algebraic_single**.

In this section, we present four algorithms with the following common pattern. Details can be found in the author's PhD thesis [10].

Common pattern. The common idea of the algorithms in [7, §3, §5, §6] is:

- (i) Based on **(GA)**, to define an algebraic set that involves a variable z_0 that corresponds to the series $F(t, a)$.
- (ii) to compute a disjunction of polynomial equations whose solution set is the algebraic set (i),
- (iii) to eliminate all variables except (z_0, t) from the polynomial characterization (ii).

We detail these 3 steps for the algorithms “duplication”, “elimination” and “geometry”. Readers can follow this section with the associated Maple worksheet available on the author's webpage, which shows how these 3 algorithms work on (1).

3.1. ALGORITHM “DUPLICATION”

Step (i). It results from **(GA)** that the following holds:

$$\forall 1 \leq i \leq k, \begin{cases} P(F(t, U_i(t)), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, U_i(t)) = 0, \\ \partial_x P(F(t, U_i(t)), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, U_i(t)) = 0, \\ \partial_u P(F(t, U_i(t)), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, U_i(t)) = 0, \end{cases}$$

and

$$\prod_{1 \leq i < j \leq k} (U_i(t) - U_j(t)) \cdot \prod_{1 \leq i \leq k} U_i(t) \cdot (U_i(t) - a) \neq 0.$$

Denote by m a new variable. In terms of polynomial equations², the above relations are equivalent to saying that the *duplicated* polynomial system $\mathcal{S}_{\text{dup}}$ defined by

$$\forall 1 \leq i \leq k, \begin{cases} P(x_i, z, t, u_i) = 0, \\ \partial_x P(x_i, z, t, u_i) = 0, \\ \partial_u P(x_i, z, t, u_i) = 0, \end{cases} \quad m \cdot \prod_{1 \leq i < j \leq k} (u_i - u_j) \cdot \prod_{1 \leq i \leq k} u_i \cdot (u_i - a) - 1 = 0,$$

admits the nontrivial solution

$$x_i = F(t, U_i(t)), \quad u_i = U_i(t), \quad z_{i-1} = \partial_u^{i-1} F(t, a), \quad \text{for } 1 \leq i \leq k. \quad (8)$$

Note that (8) uniquely determines the value associated with the variable m . Also, the polynomial system $\mathcal{S}_{\text{dup}}$ has $3k + 1$ equations and unknowns; that is, the polynomials in $\mathcal{S}_{\text{dup}}$ are seen as elements of the ring $\mathbb{Q}(t)[m, x, z, u]$. Denote by $\mathbf{V}(\mathcal{S}_{\text{dup}})$ the solution set of $\mathcal{S}_{\text{dup}}$ in $\overline{\mathbb{Q}(t)}^{3k+1}$. Assume that $\#\mathbf{V}(\mathcal{S}_{\text{dup}})$ is finite (up to deforming the equation in the spirit of

²By using the Rabinowitsch trick, instead of considering an inequation $Q \neq 0$ (for a given polynomial $Q \in \mathbb{K}[x_1, \dots, x_k]$), introduce instead an extra variable m and consider the equation $m \cdot Q - 1 = 0$ defined in $\mathbb{K}[m, x_1, \dots, x_k]$.

the proof of [9, Thm. 3], this assumption is harmless).

Step (ii). By design, the polynomial system $\mathcal{S}_{\text{dup}}$ characterizes $\mathbf{V}(\mathcal{S}_{\text{dup}})$.

Step (iii). Denote by $\mathcal{I}_{\text{dup}} \subset \mathbb{Q}(t)[m, x, z, u]$ the ideal generated by the polynomials in $\mathcal{S}_{\text{dup}}$. One applies the algorithm from Fact 3 to $\mathcal{I}_{\text{dup}}$. This algorithm outputs, for a proper choice of variable ordering, a Gröbner basis $G_{\succ_{\text{lex}}}$ of $\mathcal{I}_{\text{dup}}$ which contains an element of $\mathcal{I}_{\text{dup}} \cap \mathbb{Q}(t)[z_0]$. Applying Theorem 3 to the projection onto the z_0 -coordinate space together with Theorem 2 implies that this polynomial is nonzero and annihilates $F(t, a)$. Denote by $R \in \mathbb{Q}[t, z_0] \setminus \{0\}$ its numerator, then $R(t, F(t, a)) = 0$.

Example 1. Multiplying (1) by $(u - 1)^2$ yields $P(F(t, u), F(t, 1), \partial_u F(t, 1), t, u) = 0$, where

$$P := (u - 1)^2(1 - x + tx^3) + tu(u - 1)(2x + z_0)(x - z_0) + tu(x - z_0 - (u - 1)z_1) \in \mathbb{Q}[x, z_0, z_1, t, u].$$

Consider the polynomial system

$$\begin{aligned} \mathcal{S}_{\text{dup}} := & (P(x_1, z_0, z_1, t, u_1) = 0, \partial_x P(x_1, z_0, z_1, t, u_1) = 0, \partial_u P(x_1, z_0, z_1, t, u_1) = 0, \\ & P(x_2, z_0, z_1, t, u_2) = 0, \partial_x P(x_2, z_0, z_1, t, u_2) = 0, \partial_u P(x_2, z_0, z_1, t, u_2) = 0, \\ & m \cdot (u_1 - u_2) \cdot (u_1 - 1) \cdot (u_2 - 1) \cdot u_1 \cdot u_2 - 1 = 0), \end{aligned}$$

and let $\mathcal{I}_{\text{dup}} \subset \mathbb{Q}(t)[m, x_1, x_2, z_0, z_1, u_1, u_2]$ be the ideal generated by the polynomials in $\mathcal{S}_{\text{dup}}$.

The algorithm “duplication” computes a generator $R \in \mathcal{I}_{\text{dup}} \cap \mathbb{Q}(t)[z_0]$ and finds

$$R = (81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1)(16tz_0^2 - 8tz_0 + t - 16).$$

Let us mention that if (GA) holds but that $\#\mathbf{V}(\mathcal{S}_{\text{dup}})$ is infinite, we would then obtain the trivial polynomial $R = 0$.

3.2. ALGORITHM “ELIMINATION”

Step (i). By (6) and (GA), the system of polynomial constraints defined in $\overline{\mathbb{Q}(t)}[x, u]$ by:

$$\begin{cases} P(x, F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \\ \partial_x P(x, F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, & u(u - a) \neq 0, \\ \partial_u P(x, F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \end{cases} \quad (9)$$

admits the k solutions $(x, u) = (F(t, U_i(t)), U_i(t)) \in \overline{\mathbb{Q}(t)}^2$, for $1 \leq i \leq k$. Note that these solutions have distinct u -coordinates, i.e., the U_i are distinct. This observation can be reformulated geometrically in the following way. Consider the geometric projection $\pi : (x, u, z) \in \overline{\mathbb{Q}(t)}^{k+2} \mapsto (z) \in \overline{\mathbb{Q}(t)}^k$, and denote by $\mathcal{X} \subset \overline{\mathbb{Q}(t)}^{k+2}$ the solution set of

$$\begin{cases} P(x, z, t, u) = 0, \\ \partial_x P(x, z, t, u) = 0, & u(u - a) \neq 0, \\ \partial_u P(x, z, t, u) = 0. \end{cases} \quad (10)$$

For $\alpha \in \overline{\mathbb{Q}(t)}^k$, denote by $\#_u(\mathcal{X}, \alpha)$ the number of u -coordinates in $\pi^{-1}(\alpha) \cap \mathcal{X}$ that are not constants and define $\mathcal{F}_k(u, \mathcal{X}) := \{\alpha \in \overline{\mathbb{Q}(t)}^k \mid \#_u(\mathcal{X}, \alpha) \geq k\}$. Under assumption (GA), it holds that $(F(t, a), \dots, \partial_u^{k-1} F(t, a)) \in \mathcal{F}_k(u, \mathcal{X})$. We further assume $\#\mathcal{F}_k(u, \mathcal{X}) < +\infty$ and that for any $\alpha \in \overline{\mathbb{Q}(t)}^k$, we have $\#_u(\mathcal{X}, \alpha) < +\infty$. Again, if these assumptions do not hold,

we would obtain the trivial polynomial $R = 0$ as an output.

Step (ii). The following steps yield a polynomial characterization of the set $\mathcal{F}_k(u, \mathcal{X})$:

- (1) Compute using Theorems 2, 4 successive disjunctions of conjunctions of polynomial equations and inequations defined in $\mathbb{K}(t)[u, z]$ whose solution sets are successively the projections onto the (u, z) -coordinate space and onto the (z) -coordinate space. The union of the polynomial constraints in these disjunctions has the form $\{I, E\}$, where I is a set of inequations defined in $\mathbb{Q}(t)[u, z]$ and E is a set of equations defined in $\mathbb{Q}(t)[u, z]$. Without loss of generality, assume that the polynomials in E are a Gröbner basis of the ideal they generate, for $\succ_{\text{lex}}$ (cf. [16, Chap. 2, §2, Def. 3]) with u greater than z .
- (2) Apply Theorem 4 to E : at $z = \alpha \in \overline{\mathbb{Q}(t)}^k$ fixed, the cardinality condition in the definition of $\mathcal{F}_k(u, \mathcal{X})$ is equivalent to studying for a polynomial g_{j_0} in E :
 - if g_{j_0} has its leading coefficient in u that does not vanish at $z = \alpha$ and, by Theorem 4, if g_{j_0} is the polynomial in E of minimal degree in u that satisfies this property: checking this minimality condition yields a disjunction of polynomial equations and inequations defined in $\mathbb{Q}(t)[z]$,
 - for this g_{j_0} , it remains to add a conjunction of polynomial conditions in $\mathbb{Q}(t)[u, z]$ so that $g_{j_0}(u, z = \alpha)$ has at least k distinct solutions:
If $\deg_u(g_{j_0}) < k$, add the vanishing of all the coefficients (in u) of g_{j_0} to E ; else add the conjunction of inequations given by the non-vanishing of the $k \times k$ -minors of $M_{H_{g_{j_0}}}$ from Fact 2.
- (3) We have computed in $\mathbb{Q}(t)[u, z]$ a disjunction of conjunctions of polynomial equations and inequations whose solution set in $\overline{\mathbb{Q}(t)}^{k+1}$ is $\mathcal{F}_k(u, \mathcal{X})$. This polynomial characterization can be turned into a disjunction of conjunctions of polynomial equations (no more inequations) by using the Rabinowitsch trick in order to remove the solution set given by the inequations. The variables introduced by this trick are then eliminated using Theorem 2 and, as $\#\mathcal{F}_k(u, \mathcal{X}) < +\infty$, the so-obtained polynomials characterize the associated geometric projection (not only its Zariski closure).

Denote by $\mathcal{D}_{\text{elim}}$ the resulting disjunction of conjunctions of polynomial equations.

Step (iii). As $\#\mathcal{F}_k(u, \mathcal{X}) < +\infty$, the solution set of $\mathcal{D}_{\text{elim}}$ is finite; we therefore proceed as in Section 3.1. We can thus apply the algorithm from Fact 3 to each conjunction in $\mathcal{D}_{\text{elim}}$. Each application of this algorithm allows one to compute a nonzero polynomial of $\mathbb{Q}(t)[z_0]$: this polynomial might be equal to 1 when there is no solution to the studied conjunction. Consider the numerators of all these polynomials and denote by $R \in \mathbb{Q}[t, z_0]$ their product. By [7, Prop. 5.5], we have $R(t, F(t, a)) = 0$.

Observe that, all in all, the main computation of this algorithm consists in computing a Gröbner basis of $\{P, \partial_x P, \partial_u P, m \cdot u \cdot (u - a) - 1\}$ for a monomial order with $m > x > u > z$.

Example 1 (cont.). Denote $\mathcal{S} := \{P, \partial_x P, \partial_u P, m \cdot u(u - 1) - 1\}$ and $\mathcal{X} \subset \overline{\mathbb{Q}(t)}^4$ the solution set of the conditions $P = 0 \wedge \partial_x P = 0 \wedge \partial_u P = 0 \wedge u(u - 1) \neq 0$ in $\overline{\mathbb{Q}(t)}^4$.

The goal is to characterize $\mathcal{F}_k(u, \mathcal{X}) := \{\alpha \in \overline{\mathbb{Q}(t)}^2 \mid \#_u(\mathcal{X}, \alpha) \geq 2\}$ as the solution set of some explicit polynomial constraints and deduce $R \in \mathbb{Q}[t, z_0] \setminus \{0\}$ such that $R(t, F(t, 1)) = 0$.

The algorithm “elimination”:

- (1) characterizes the projection of $\mathcal{X}$ onto the (u, z_0, z_1) -coordinate space:

```
G := Groebner[Basis](S, lexdeg([m, x], [u, z0, z1, t])):
E := Groebner[Basis](remove(has, G, {m, x}), lexdeg([u], [z0, z1, t])):
Leading := select(has, G, {m, x}):
```

The first element of Leading has the form

$$1534(u-1) \cdot x + (\text{polynomial in } u, z_0, z_1, t).$$

As $\mathcal{X} \cap \{u = 1\} = \emptyset$, it results from the extension theorem that the zero set of E in $\overline{\mathbb{Q}(t)}^3$ is precisely the projection of $\mathcal{X}$ onto the (u, z_0, z_1) -coordinate space.

- (2) identifies algebraic conditions in E describing the cardinality of the fiber in $\mathcal{F}_k(u, \mathcal{X})$:

```
# Dealing with the polynomials of degree < 2 in u in E:
NewConditions := [];
for i to nops(E) do
  if degree(E[i], u) < 2 then
    NewConditions := [op(NewConditions), coeffs(E[i], u)]:
  fi:
od:
E := Groebner[Basis]([op(E), op(NewConditions)],
  lexdeg([u], [z0, z1, t])):
# Dealing with the polynomials of degree > 1 in u in E:
map(p -> degree(p, u), E);
# prints the degrees in u of the polynomials in E
```

```
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3]
```

In order to have $\#_u(\mathcal{X}, \alpha) \geq 2$ for a given $\alpha \in \overline{\mathbb{Q}(t)}^2$, we shall exclude the solution set of $\text{disc}_u(E[10])$:

```
sat := discrim(E[10], u):
```

Denote by $c_{10}(z_0, z_1) \in \mathbb{Q}(t)[z_0, z_1]$ the coefficient of u^2 in $E[10]$. As $\mathbf{V}(E) \cap \{c_{10} = 0\} = \emptyset$, the algorithm deduces from Theorem 4 that $\mathcal{F}_k(u, \mathcal{X})$ is the solution set of the polynomial equations given by E and of the inequation $\text{sat} \neq 0$.

- (3) turns the inequation $\text{sat} \neq 0$ into the equation $m \cdot \text{sat} - 1 = 0$ by introducing an extra variable m . Finally, it deduces a nonzero polynomial $R \in \mathbb{Q}[t, z_0]$ annihilating $F(t, 1)$.

```
Delim := remove(has,
  Groebner[Basis]([op(E), m*sat-1], lexdeg([m], [u, z1, z0, t])), {m}):
H := Groebner[Basis](Delim, lexdeg([u, z1], [z0, t])):
R := factor(op(remove(has, H, {u, z1})))
```

One finds

$$R = (16tz_0^2 - 8tz_0 + t - 16)(81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1).$$

3.3. ALGORITHM “GEOMETRY”

Step (i). By (6) and (GA), the system of polynomial constraints defined in $\overline{\mathbb{Q}(t)}[x, z_1, u]$ by

$$\begin{cases} P(x, F(t, a), z_1, \partial_u^2 F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \\ \partial_x P(x, F(t, a), z_1, \partial_u^2 F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, & u(u - a) \neq 0, \\ \partial_u P(x, F(t, a), z_1, \partial_u^2 F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \end{cases} \quad (11)$$

has the k distinct solutions $(x, z_1, u) = (F(t, U_i(t)), \partial_u F(t, a), U_i(t)) \in \overline{\mathbb{Q}(t)}^3$, for $1 \leq i \leq k$.

Denote by $\tilde{z}_1$ the set of variables $z_0, z_2, \dots, z_{k-1}$. Consider the geometric projection $\pi_{\tilde{z}_1} : (x, u, z) \in \overline{\mathbb{Q}(t)}^{k+2} \mapsto (\tilde{z}_1) \in \overline{\mathbb{Q}(t)}^{k-1}$ and define

$$\mathcal{S}_k(\mathcal{X}) := \{\alpha = (\alpha_0, \dots, \alpha_{k-1}) \in \overline{\mathbb{Q}(t)}^k \mid \alpha \in \pi(\mathcal{X}) \wedge \# \mathcal{X} \cap \pi_{\tilde{z}_1}^{-1}((\alpha_0, \alpha_2, \alpha_3, \dots, \alpha_{k-1})) \geq k\}.$$

Under assumption (GA), it holds that $(F(t, a), \dots, \partial_u^{k-1} F(t, a)) \in \mathcal{S}_k(\mathcal{X})$. The algorithm from [7, §6] requires many additional technical assumptions (see the beginning of [7, §6]): among them is the finiteness of $\#\mathcal{S}_k(\mathcal{X})$. We do not make the other assumptions explicit for the sake of simplicity. In the rest of this section, denote by $\succ_{\text{bgrvlex}}$ a block monomial order over $\mathbb{Q}(t)[m, x, u, z]$ defined as follows:

- We use the graded monomial order $\succ_{\text{grvlex}}$ (cf. [16, Chap. 2, §2, Def. 6]) on each of the two blocks $\{m, x, u, z_1\}$ and $\{\tilde{z}_1\}$,
- Two monomials are compared w.r.t. the variables m, x, u, z_1 . In case of equality, they are compared w.r.t. the variables $\tilde{z}_1$.

Example 2. For $k = 3$:

$$x^3 u z_1^2 m \succ_{\text{bgrvlex}} z_0^{20} z_2^3, \quad x^3 u z_1^2 m z_2 \succ_{\text{bgrvlex}} x^3 u z_1^2 m z_0 \quad \text{and} \quad z_0^4 \succ_{\text{bgrvlex}} z_0^2 z_2.$$

Under these assumptions and the fact that $\mathcal{S}_k(\mathcal{X})$ is a finite set (else, the algorithm would return the trivial polynomial $R = 0$), the algorithm designed in [7, §6] works as follows.

Step (ii). The following steps yield a characterization of the set $\mathcal{S}_k(\mathcal{X})$:

- (1) Compute a Gröbner basis G of the ideal $\mathcal{I} := \langle P, \partial_x P, \partial_u P, m \cdot u(u - a) - 1 \rangle \subset \mathbb{Q}(t)[m, x, u, z]$ for the order $\succ_{\text{bgrvlex}}$.
- (2) Compute, using normal form computations modulo $\mathcal{I}^3$, the matrix M_{z_1} of the multiplication map m_{z_1} in the quotient ring $\mathbb{Q}(t, \tilde{z}_1)[m, x, u, z_1]/j(\mathcal{I})$, where $j(\mathcal{I})$ is the image of the injective map $j : \mathbb{Q}(t)[m, x, u, z] \rightarrow \mathbb{Q}(t, \tilde{z}_1)[m, x, u, z_1]$.
- (3) Compute the characteristic polynomial $\chi_{z_1} \in \mathbb{Q}(t, \tilde{z}_1)[T]$. Using the assumptions of [7, §6], we can replace χ_{z_1} by its numerator with respect to the variables $\tilde{z}_1$, so that we now have $\chi_{z_1} \in \mathbb{Q}(t)[\tilde{z}_1, T]$.
- (4) Denote by $\mathcal{D}_{\text{geom}}$ the set of equations given by the vanishing of the polynomials in G and of the new conditions $(\chi_{z_1})|_{T=z_1} = 0 \wedge \dots \wedge (\partial_T^{k-1} \chi_{z_1})|_{T=z_1} = 0$. The solution set of $\mathcal{D}_{\text{geom}}$ is $\mathcal{S}_k(\mathcal{X})$.

Step (iii). As the solution set $\mathcal{S}_k(\mathcal{X})$ of $\mathcal{D}_{\text{geom}}$ is finite, we perform this step in the same spirit as we did in Section 3.1. We thus apply the algorithm from Fact 3 to the ideal generated by the polynomials associated with the equations in $\mathcal{D}_{\text{geom}}$. This algorithm outputs a nonzero polynomial of $\mathbb{Q}(t)[z_0]$. Consider the numerator of this polynomial and denote it by $R \in \mathbb{Q}[t, z_0]$. By [7, Prop. 6.4], we have $R(t, F(t, a)) = 0$.

³Normal form computations modulo ideals are generalizations of Euclidean divisions by polynomials.

Example 1 (cont.). The goal is to characterize $S_k(\mathcal{X})$ as the solution set of some polynomial constraints and deduce a nonzero polynomial $R \in \mathbb{Q}[t, z_0]$ such that $R(t, F(t, 1)) = 0$.

The algorithm “geometry”:

- (1) computes a Gröbner basis of the ideal generated by S for the order $\succ_{\text{bgrexlex}}$:
 $G := \text{Groebner}[\text{Basis}](S, \text{lexdeg}([m, x, u, z_1], [t, z_0]))$;
- (2) computes the matrix M_{z_1} of the multiplication map $m_{z_1} : f \mapsto z_1 \cdot f$ in the quotient ring $\mathbb{Q}(t, z_0)[m, x, u, z_1]/\langle j(S) \rangle$, where j is the usual inclusion map of $\mathbb{Q}(t)[m, x, u, z_0, z_1]$ in $\mathbb{Q}(t, z_0)[m, x, u, z_1]$:
 $L_1, L_2 := \text{Groebner}[\text{NormalSet}](\text{subs}(t = \text{rand}(), z_0 = \text{rand}()), G, \text{tdeg}(m, x, u, z_1))$;
 $M := \text{Groebner}[\text{MultiplicationMatrix}](z_1, L_1, L_2, G, \text{tdeg}(m, x, u, z_1))$;
- (3) computes the characteristic polynomial of M_{z_1} and defines χ_{z_1} to be its numerator:
 $\text{xi_z1} := \text{LinearAlgebra}[\text{CharacteristicPolynomial}](M, T)$;
 $\text{chi_z1} := \text{factor}(\text{numer}(\text{xi_z1}))$;
 $\text{Dgeom} := \text{Groebner}[\text{Basis}](\text{op}(G), \text{chi_z1}, \text{diff}(\text{chi_z1}, T), \text{tdeg}(m, x, u, z_1, t, z_0))$;
- (4) deduces that $S_k(\mathcal{X})$ is the solution set of the constraints $P = 0 \wedge \partial_x P = 0 \wedge \partial_u P = 0 \wedge u(u - 1) \neq 0 \wedge \text{disc}_T(\chi_{z_1}) = 0$. An annihilating polynomial of $F(t, 1)$ is given by the discriminant $R \in \mathbb{Q}[t, z_0]$ of χ_{z_1} with respect to T , that is

$$R = 12230590464t^{14} \cdot (16tz_0^2 - 8tz_0 + t - 16)^2 \cdot (1 + (3z_0 - 1)^2t^2 + (-6z_0 - 14)t)^3 \cdot (tz_0 + 1)^{12} \cdot (81t^2z_0^3 - 9t(9t - 2)z_0^2 + (27t^2 - 66t + 1)z_0 - 3t^2 + 47t - 1)^2.$$

3.4. ALGORITHM “HYBRID”

The hybrid guess-and-prove algorithm [7, §4]:

- (1) precomputes bounds b_{z_0}, b_t such that $\deg_t(R) \leq b_t$ and $\deg_{z_0}(R) \leq b_{z_0}$, for some nonzero polynomial $R \in \mathbb{Q}[t, z_0]$ annihilating $F(t, a)$,
- (2) computes the truncated series $F(t, a) \bmod t^{2b_t b_{z_0} + 1}$ by using a Newton-like algorithm on (2),
- (3) relying on Hermite-Padé approximants, guesses a polynomial $M \in \mathbb{Q}[t, z_0]$ such that $M(t, F(t, a)) = O(t^{(b_t+1)(b_{z_0}+1)-1})$,
- (4) checks that $M(t, F(t, a)) = O(t^{2b_t b_{z_0} + 1})$.

When step 4 is satisfied, the argument based on the use of resultants in [6, §2.2.2] demonstrates that $M(t, F(t, a)) = 0$.

Example 1 (cont.). We compute a polynomial $R \in \mathbb{Q}[t, z_0] \setminus \{0\}$ such that $R(t, F(t, 1)) = 0$.

The algorithm “hybrid”:

- (1) draws uniformly at random a prime number p and some $\theta \in \mathbb{F}_p$, say $p = 12301$ and $\theta = 1328$, and applies the algorithm “elimination” with S replaced by $(S)|_{t=\theta}$ and $\mathbb{Q}$ replaced by $\mathbb{F}_p$. It outputs

$$7957(z_0 + 11829)(z_0^3 + 4863z_0^2 + 8711z_0 + 3012)(z_0 + 6622)$$

The degree being 5, it sets $b_{z_0} = 5$. Similarly, it obtains $b_t = 3$.

- (2) computes the 32 first terms of $F(t, 1)$:

$$F_1 = 1 + t + 6t^2 + 54t^3 + \cdots + 913075994651156584840651326232625946t^{31} \bmod t^{32}$$

- (3) uses the function `seriestoalgeq` of the Maple package `gfun` [25]:

```
Rguess := gfun[seriestoalgeq](series(F1, t, degree(F1, t)), T(t))[1]:
collect(subs(T(t) = z0, Rguess), z0, factor);
```

$$81t^2z_0^3 - 9t(9t-2)z_0^2 + (27t^2-66t+1)z_0 - 3t^2 + 47t - 1$$

The above polynomial, denoted by M , is a candidate for annihilating $F(t, 1)$.

(4) proves the guessed polynomial with the below lines

```
series(subs(z0 = F1, Rguess), t, 31);
```

$$O(t^{31})$$

$$\text{Thus } R = 81t^2z_0^3 - 9t(9t-2)z_0^2 + (27t^2-66t+1)z_0 - 3t^2 + 47t - 1.$$

4. DETAILS ON THE IMPLEMENTATIONS

4.1. PARADIGMS USED IN THE IMPLEMENTATIONS

As we target practical efficiency, the implementations of the options “duplication”, “elimination” and “geometry” incorporate the following key improvements from computer algebra.

Evaluation–interpolation. To reduce computations over $\mathbb{Q}(t)$ to computations over $\mathbb{Q}$, we perform evaluation–interpolation on t .

Multi-modular arithmetic. We rely on fast multi-modular arithmetic. This consists in applying multiple times a given algorithm with the base field $\mathbb{Q}$ replaced by successive distinct prime fields $\mathbb{F}_p$. These computations output various images mod p of the same $R \in \mathbb{Q}[t, z_0]$. From there, one lifts the modular coefficients over $\mathbb{Q}$ by applying the Chinese Remainder Theorem (CRT) together with rational number reconstruction [19, §5.10].

Random specializations in “hybrid”. Computing the bounds (b_t, b_{z_0}) in step 1 is done using the option “elimination” of Section 3.2. More precisely, computing b_{z_0} is done by specializing t at a random value θ of some random prime field $\mathbb{F}_p$ and performing the option “elimination” with base field $\mathbb{F}_p$: it outputs $R(\theta, z_0)$ and yields $b_{z_0} = \deg_{z_0}(R(\theta, z_0))$ (a similar computation is done for computing b_t). The generation of terms is done using a divide and conquer approach which computes the first terms of $F(t, u) \bmod t^{2b_t b_{z_0} + 1}$ before specializing them to $u = a$. Finally, the guessing is done by computing Hermite-Padé approximants. For this, we use the function `seriestoalgeq` of the Maple package `gfun` [25].

4.2. GETTING STARTED

4.2.1. Installation and customization. Files provided: *DDE-Solver* is available in the “.mla” format. If one wishes to customize the package, the Maple scripts “ddesolver.mpl” and “build.mpl” can be downloaded on the dedicated [github webpage](#). Downloading these two scripts lets one customize the package: edit “ddesolver.mpl”, then run “build.mpl” to regenerate “ddesolver.mla”.

Customization of *DDE-Solver*: Replacing “PATH/TO” in the Maple file “build.mpl” by the relevant path to the directory where “ddesolver.mpl” and “build.mpl” are located allows one, by executing “build.mpl”, to generate a new version of “ddesolver.mla”.

Loading DDE-Solver: The Maple variable `libname` must be set so that Maple can find “dde-solver.mla”.

```
libname := "/home/yourname/ddesolver/lib", libname:
```

Once `libname` has been correctly set up, one executes in Maple

```
with(ddesolver);
```

in order to load and use the package.

4.2.2. Using DDE-Solver. The arguments of `algebraic_single` are (P, k, var) , where P is the polynomial in (3), k is the order of the DDE and var is $[x, z, t, u]$ (precisely in that order).

The output of `algebraic_single` (P, k, var) is some nonzero polynomial $R \in \mathbb{Q}[t, z_0]$ such that $R(t, F(t, a)) = 0$.

Example 3. Consider (1). Multiplying (1) by $(u - 1)^2$ yields

$$0 = (u - 1)^2(1 - F(t, u) + tuF(t, u)^3) + tu(u - 1)(2F(t, u) + F(t, 1))(F(t, u) - F(t, 1)) \\ + tu(F(t, u) - F(t, 1) - (u - 1)\partial_u F(t, 1)).$$

Thus $P := (u - 1)^2(1 - x + tuu^3) + tu(u - 1)(2x + z_0)(x - z_0) + tu(x - z_0 - (u - 1)z_1)$ and $k := 2$. We now solve it with Maple

```
P := (u-1)^2*(1-x+tu*u^3)+tu*(u-1)*(2*x+z0)*(x-z0)+tu*(x-z0-(u-1)*z1);
with(ddesolver):
algebraic_single(P, 2, [x, z0, z1, t, u]);
```

outputs

$$(16tz_0^2 - 8tz_0 + t - 16) * \\ (81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1)$$

Thus $R := (16tz_0^2 - 8tz_0 + t - 16)(81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1)$ is an annihilating polynomial of $F(t, 1)$. A direct analysis on the solutions of $R(t, z_0)$ that are finite at $t = 0$ shows that the second factor of R is the minimal polynomial of $F(t, 1)$.

4.2.3. Options. It is possible to benefit in practice from two options: the choice of the *algorithm*, and the choice of the *variable* (either t or z_0) on which we perform evaluation–interpolation.

Choice of the algorithm. Four algorithms are implemented: “duplication”, “elimination”, “geometry” and “hybrid”. By default, the algorithm used is “elimination”.

Choice of the variable. The two choices on which we perform the evaluation–interpolation strategy are t and z_0 . The default choice is t . This option affects the timings (see Figure 2). Both choices yield the same output.

The choice of the algorithm and of the variable on which we perform the evaluations is made by executing

```
algebraic_single(P, k, var, algorithm, variable).
```

Note that these two options must either both be omitted or both be given.

Example 4. Continuing Example 3 with algorithm = “geometry” and variable = z_0 yields

```
algebraic_single(P, 2, [x, z0, z1, t, u], "geometry", z0);
```

$$(16*t*z_0^2-8*t*z_0+t-16) * \\ (81*t^2*z_0^3-81*t^2*z_0^2+27*t^2*z_0+18*t*z_0^2-3*t^2-66*t*z_0+47*t+z_0-1)$$

We refer to the next section for a discussion on the choice of the parameter variable.

4.3. EXAMPLES

4.3.1. Impact of the option variable. The motivation of this study is the following. Assume that we perform evaluation–interpolation on z_0 . The chosen algorithm (e.g. “elimination” from Section 3.2) will be run a first time with t specialized (in order to get the degree in z_0 of the output $R \in \mathbb{Q}[t, z_0]$), then it will be run multiple times with z_0 specialized (in order to evaluate–interpolate R w.r.t z_0). Hence if the algorithm is faster at t specialized than at z_0 specialized (and if the partial degrees of the output R are “close”), then the choice of the evaluation–interpolation variable has a significant impact on the timings as we now show.

Example 5. Consider the DDE associated with the enumeration of intervals in the 3-Tamari lattices [8, Prop. 8]:

$$F(t, u) = u + tuF(t, u) \frac{F(t, u) \frac{F(t, u) - F(t, 1)}{u-1} - F(t, 1) \partial_u F(t, 1)}{u-1} - \frac{F(t, 1)^2 \partial_u^2 F(t, 1)}{2} - F(t, 1) \partial_u F(t, 1)^2, \quad (12)$$

Multiplying (12) by $(u-1)^3$ yields the functional equation

$$P(F(t, u), F(t, 1), \partial_u F(t, 1), \partial_u^2 F(t, 1), t, u) = 0,$$

where $P := -tu^3xz_0^2z_2 - 2tu^3xz_0z_1^2 - 2tu^2x^2z_0z_1 + 2tu^2xz_0^2z_2 + 4tu^2xz_0z_1^2 + 2tux^4 - 2tux^3z_0 + 2tux^2z_0z_1 - tuxz_0^2z_2 - 2tuxz_0z_1^2 + 2u^4 - 2u^3x - 6u^3 + 6u^2x + 6u^2 - 6ux - 2u + 2x \in \mathbb{Q}[x, z_0, z_1, z_2, t, u]$. We thus have in Maple

```
P := -t*u^3*x*z0^2*z2-2*t*u^3*x*z0*z1^2-2*t*u^2*x^2*z0*z1
      +2*t*u^2*x*z0^2*z2+4*t*u^2*x*z0*z1^2+2*t*u*x^4-2*t*u*x^3*z0
      +2*t*u*x^2*z0*z1-t*u*x*z0^2*z2-2*t*u*x*z0*z1^2+2*u^4
      -2*u^3*x-6*u^3+6*u^2*x+6*u^2-6*u*x-2*u+2*x:
with(ddesolver):
time(algebraic_single(P, 3, [x, z0, z1, z2, t, u]));
      # variable = t (implicitly chosen)

2089.599

time(algebraic_single(P, 3, [x, z0, z1, z2, t, u],
      "elimination", z0)); # variable = z0

141.897

R := algebraic_single(P, 3, [x, z0, z1, z2, t, u],
      "elimination", z0);
R := t^5*z0^16+135*t^4*z0^13+1024*t^4*z0^12+7290*t^3*z0^10
      -1762560*t^3*z0^9+393216*t^3*z0^8+196830*t^2*z0^7+111694464*t^2*z0^6
      +580976640*t^2*z0^5+t*(67108864*t+2657205)*z0^4-661978656*t*z0^3
      +4721836032*t*z0^2+(-8371830784*t+14348907)*z0+4294967296*t-14348907
```

4.3.2. *Practical results.* Figure 2 gathers some timings obtained after using the function `algebraic_single`.

Each column is associated with a DDE taken from the literature (cf. [4, 8, 9, 13]). For each DDE, we specify: its order k , the variable on which is performed the evaluation–interpolation, the algorithm used and the bidegree of the output $R \in \mathbb{Q}[t, z_0]$.

From Figure 2, we reinforce the message of Section 4.3.1 and draw some conclusions:

- *Choice of the algorithm.* The best algorithm to choose depends in an important way on the DDE. Even among DDEs of the same family (e.g. DDEs [4], [5] and [8]) the answer is not clear: the first one is solved faster with algorithm = “geometry” while the last two are solved faster with algorithm = “hybrid”. The efficiency of this last option algorithm = “hybrid” comes from the fact that the output polynomial has small bidegree (3, 9), thus allowing a fast computation of the first terms of $F(t, 1)$; and at the same time, that the computations are hard with the other algorithms.

This difficulty to predict which algorithm is more efficient for an input DDE justifies letting the user choose the algorithm which shall be applied.

- *Choice of the variable.* As in Section 4.3.1, the table shows that the choice of the variable (z_0 or t) on which we perform evaluation–interpolation is important. Let us pick one example to illustrate it. For the DDE [3], performing evaluation–interpolation over t with algorithm = “geometry” takes 2 minutes while it takes 54 minutes when performing evaluation–interpolation over z_0 . With the notations of Section 3.3, the reason is that at z_0 specialized to some random $\theta \in \mathbb{F}_p$, the main computational cost (in addition to the Gröbner basis computation) is to compute the discriminant w.r.t. T of $(\chi_{z_1})|_{z_0=\theta} \in \mathbb{Q}[t, T]$, whose partial degrees are 6 in T and 423 in t . At the same time at t specialized to some random $\nu \in \mathbb{F}_p$, the polynomial $(\chi_{z_1})|_{t=\nu} \in \mathbb{Q}[z_0, T]$ has degree 15 in z_0 and 6 in T : it yields a faster computation of its discriminant w.r.t. T .

As all these intermediate computational data are hard to predict from the input DDE, we let the user choose on which variable the evaluation–interpolation shall be performed.

- *Solving DDEs previously out of reach.* The DDEs [7], [8] are solved via theoretical arguments (cf. [12, 13]). Here, it is the first time that they are solved in a “click”.

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Data	[1]	[2]	[3]	[4]	[4b]	[5]	[6]	[7]	[8]
k	2	2	2	2	2	3	3	3	4
eval–interp variable	z_0 t	z_0 t	z_0 t	z_0 t	z_0 t	z_0 t	z_0 t	z_0 t	z_0 t
“duplication”	0.5s 0.7s	1.6s 2.3s	41m* 10m*	2.7s 1.5s	11s 12s	2m45s 2m50s	13h 27h	∞ ∞	∞ ∞
“elimination”	0.2s 0.3s	0.4s 0.4s	1h20m* 4m*	1.7s 1.2s	10s 9s	1m10s 45s	2m20s 35m	1m 7m30s	2d19h 2d
“geometry”	2s 2.2s	1.6s 0.9s	54m* 2m*	0.8s 0.7s	6m19s 5m54s	$\times$ $\times$	$\times$ $\times$	$\times$ $\times$	$\times$ $\times$
“hybrid”	50s	2m	∞	1.4s	∞	18s	1h42m	34s	2h41m
$(\deg_t(R), \deg_{z_0}(R))$	(4, 9)	(2, 3)	(132, 6)	(3, 5)	(14, 23)	(3, 7)	(5, 16)	(2, 4)	(3, 9)

FIGURE 2. Practical results of **algebraic_single** on DDEs from the literature.

Computations are done in Maple on a computer equipped with Intel® Xeon® Gold CPU 6246R v4 @ 3.40GHz and 1.5TB of RAM with 1 thread. Gröbner bases computations are performed using the C library **msolve** [5], implemented by Berthomieu, Eder and Safey El Din. Maple version 2024.

- [1]: Enumeration of intervals in the 2-Tamari lattices, [8, Prop. 8] for $m = 3$,
- [2]: Enumeration of intervals in the 2-greedy Tamari posets, [13, Prop. 3.1] for $m = 2$,
- [3]: Enumeration of non-separable near-triangulations in which all internal vertices have degree at least 5, [4, Prop. 4.3, Eq. 22],
- [4]: Enumeration of 3-constellations, [9, Equation 29],
- [4b]: DDE of 3-constellations with $2F(t, u)$ replaced by $3F(t, u)$, [9, Equation 29],
- [5]: Enumeration of 4-constellations, [9, Prop. 12] for $m = 4$,
- [6]: Enumeration of intervals in the 3-Tamari lattices, [8, Prop. 8] for $m = 4$,
- [7]: Enumeration of intervals in the 3-greedy Tamari posets, [13, Prop. 3.1] for $m = 3$,
- [8]: Enumeration of 5-constellations. [9, Prop. 12] for $m = 5$.

- s, m, h, d: seconds, minutes, hours, days.
- ∞ : The computations did not finish within 5 days,
- $\times$: The algorithm is not implemented for $k > 2$,
- *: We added the condition $m' \cdot (1 + t^3)(1 - t^3)t - 1 = 0$ to the input polynomial constraints.

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Email address: notarantonio@lipn.univ-paris13.fr